

End to End Fleet Analytics & Route Profitability Engine

Block 1:

In the first block of code, we load the core datasets, core datasets are loads.csv, trips.csv, routes.csv, fuel_purchases.csv, maintenance_records.csv.

Little Explanation of the datasets,
loads.csv looks like this:

load_id	customer_id	route_id	load_date	load_type	weight_lbs	pieces	revenue	fuel_surcharge	accessorial_charges	load_status	booking_type
LOAD00000001	CUST00183	RTE00019	01-01-2022	Dry Van	19178	13	3045.23	406.72	100	Completed	Spot
LOAD00000002	CUST00076	RTE00058	01-01-2022	Dry Van	27761	22	1224.48	98.61	0	Completed	Dedicated
LOAD00000003	CUST00027	RTE00048	01-01-2022	Refrigerated	35594	16	7171.12	792.88	0	Completed	Spot
LOAD00000004	CUST00088	RTE00013	01-01-2022	Refrigerated	33274	10	1308.2	141.33	50	Completed	Spot
LOAD00000005	CUST00185	RTE00020	01-01-2022	Dry Van	40257	10	3317.18	738.48	0	Completed	Spot
LOAD00000006	CUST00193	RTE00004	01-01-2022	Refrigerated	34671	10	3582.78	383.23	75	Completed	Spot
LOAD00000007	CUST00169	RTE00038	01-01-2022	Dry Van	14530	4	3872.46	314.64	150	Completed	Contract
LOAD00000008	CUST00049	RTE00016	01-01-2022	Refrigerated	16927	26	5914.9	463.93	100	Completed	Spot
LOAD00000009	CUST00075	RTE00025	01-01-2022	Refrigerated	38964	3	4759.42	565.38	150	Completed	Dedicated

trips.csv looks like this:

	A	B	C	D	E	F	G	H	I	J	K	L
1	trip_id	load_id	driver_id	truck_id	trailer_id	dispatch_date	actual_distance_miles	actual_duration_hours	fuel_gallons_used	average_mpg	idle_time_hours	trip_status
2	TRIP00000001	LOAD00000001	DRV00117	TRK00035	TRL00167	01-01-2022	1314	26.2	183.8	7.15	3.5	Completed
3	TRIP00000002	LOAD00000002	DRV00141	TRK00108	TRL00082	01-01-2022	515	8.6	93.6	5.5	8.3	Completed
4	TRIP00000003	LOAD00000003	DRV00032	TRK00031	TRL00138	01-01-2022	2509	45	339.1	7.4	12	Completed
5	TRIP00000004	LOAD00000004	DRV00083	TRK00105	TRL00018	01-01-2022	717	11.1	110.3	6.5	9.6	Completed
6	TRIP00000005	LOAD00000005	DRV00044	TRK00076	TRL00054	01-01-2022	2243	35	328.9	6.82	11.6	Completed
7	TRIP00000006	LOAD00000006	DRV00120	TRK00062	TRL00007	01-01-2022	1977	39.4	358.8	5.51	6.6	Completed
8	TRIP00000007	LOAD00000007	DRV00055	TRK00002	TRL00093	01-01-2022	1350	20.9	230.8	5.85	2.3	Completed
9	TRIP00000008	LOAD00000008	DRV00065	TRK00035	TRL00148	01-01-2022	2748	50.5	437.6	6.28	10.5	Completed
10	TRIP00000009	LOAD00000009	DRV00129	TRK00014	TRL00083	01-01-2022	3226	51.4	465.5	6.93	4.1	Completed
11	TRIP00000010	LOAD00000010	DRV00029	TRK00116	TRL00126	01-01-2022	3388	60.2	460.3	7.36	5.7	Completed
12	TRIP00000011	LOAD00000011	DRV00132	TRK00085	TRL00083	01-01-2022	538	9.9	72.2	7.45	4.9	Completed
13	TRIP00000012	LOAD00000012	DRV00088	TRK00039	TRL00142	01-01-2022	861	14.3	150.3	5.73	6.7	Completed
14	TRIP00000013	LOAD00000013	DRV00070	TRK00086	TRL00119	01-01-2022	689	12.4	112	6.15	3.6	Completed

routes.csv looks like this:

	A	B	C	D	E	F	G	H	I
1	route_id	origin_city	origin_state	destination_city	destination_state	typical_distance_miles	base_rate_per_mile	fuel_surcharge_rate	typical_transit_days
2	RTE00001	Atlanta	GA	Chicago	IL	677	1.7	0.19	1
3	RTE00002	Atlanta	GA	Miami	FL	697	2.08	0.22	1
4	RTE00003	Chicago	IL	Los Angeles	CA	2003	2.55	0.29	3
5	RTE00004	Chicago	IL	Portland	OR	2017	1.84	0.19	3
6	RTE00005	Chicago	IL	Las Vegas	NV	1749	2.41	0.26	3
7	RTE00006	Chicago	IL	Charlotte	NC	675	2.13	0.28	1
8	RTE00007	Dallas	TX	New York	NY	1576	2.54	0.32	2
9	RTE00008	Dallas	TX	Denver	CO	762	2.62	0.15	1
10	RTE00009	Dallas	TX	Indianapolis	IN	878	1.95	0.17	1
11	RTE00010	New York	NY	Philadelphia	PA	92	1.61	0.17	1
12	RTE00011	New York	NY	Columbus	OH	547	1.69	0.24	1
13	RTE00012	Phoenix	AZ	Philadelphia	PA	2389	2.71	0.19	4
14	RTE00013	Phoenix	AZ	Denver	CO	673	1.7	0.21	1
15	RTE00014	Phoenix	AZ	Kansas City	MO	1204	2.14	0.19	2

fuel_purchases.csv looks like this:

	A	B	C	D	E	F	G	H	I	J	K
1	fuel_purchase_id	trip_id	truck_id	driver_id	purchase_date	location_city	location_state	gallons	price_per_gallon	total_cost	fuel_card_number
2	FUEL00000001	TRIP00051284	TRK00045	DRV00102	22-10-2023 05:00	Columbus	MN	131.6	3.399	447.31	FC567161
3	FUEL00000002	TRIP00073723	TRK00013		04-08-2024 08:00	New York	AZ	139.9	3.18	444.88	FC717910
4	FUEL00000003	TRIP00018286	TRK00024	DRV00142	23-08-2022 13:00	Seattle	NE	189.3	3.804	720.1	FC912816
5	FUEL00000004	TRIP00008465	TRK00018	DRV00047	20-04-2022 00:00	Memphis	NY	130.7	3.579	467.78	FC776357
6	FUEL00000005	TRIP00069700	TRK00054		13-06-2024 09:00	Columbus	AZ	185.3	3.422	634.1	FC391024
7	FUEL00000006	TRIP00062650	TRK00012	DRV00104	18-03-2024 23:00	Dallas	NY	182.2	3.494	636.61	FC277504
8	FUEL00000007	TRIP00000851	TRK00072	DRV00127	11-01-2022 21:00	Miami	TX	146.3	3.914	572.62	FC256663
9	FUEL00000008	TRIP00012033	TRK00073	DRV00084	06-06-2022 00:00	Phoenix	PA	113.3	4.471	506.56	FC543608
10	FUEL00000009	TRIP00023596	TRK00101	DRV00080	28-10-2022 02:00	Miami	NC	127	4.445	564.51	FC484857

maintenance_records.csv. looks like this:

	A	B	C	D	E	F	G	H	I	J	K	L
1	maintenance_id	truck_id	maintenance_date	maintenance_type	odometer_reading	labor_hours	labor_cost	parts_cost	total_cost	facility_location	downtime_hours	service_description
2	MAINT00000001	TRK00085	01-01-2022	Inspection	400255	7.8	781.42	10.41	791.83	Kansas City	22.2	Emergency Inspection
3	MAINT00000002	TRK00041	01-01-2022	Tire	268041	0.9	80.91	3207.16	3288.07	Seattle	8	Scheduled Tire
4	MAINT00000003	TRK00090	01-01-2022	Preventive	698915	7.2	733.18	1570.57	2303.75	Miami	16.5	Routine Preventive
5	MAINT00000004	TRK00056	02-01-2022	Inspection	416522	2.8	298.8	67.48	366.28	Kansas City	34.6	Emergency Inspection
6	MAINT00000005	TRK00029	02-01-2022	Repair	179968	4.6	409.08	982.59	1391.67	Omaha	4.4	Emergency Repair
7	MAINT00000006	TRK00120	02-01-2022	Preventive	680507	5	498.12	3029.54	3527.66	Omaha	43.5	Scheduled Preventive
8	MAINT00000007	TRK00017	02-01-2022	Repair	599078	0.9	83.38	1098.9	1182.28	Denver	34.8	Emergency Repair
9	MAINT00000008	TRK00038	03-01-2022	Tire	605379	7.1	785.79	1483.25	2269.04	Miami	3.9	Scheduled Tire
10	MAINT00000009	TRK00054	03-01-2022	Inspection	232147	4	371.51	143.21	514.72	Dallas	14	Routine Inspection

Block 2:

In the second block,

1. Standardize date columns of loads_df, trips_df, main_df and fuel_df.
2. Handle the missing values explicitly. If trips.csv/ fuel_purchases.csv has no driver assigned, we fill it with 'UNASSIGNED', so joins don't break.
3. Drop completely duplicate rows if any exist, in any csv.

Data Cleaning is complete, Data columns standardized.

Block 3:

In the third block, we do feature engineering.

1. First Feature is total load revenue, which is the sum of revenue, fuel surcharge, and accessory charges.
2. Second Feature is aggregated fuel cost per trip. One trip can have multiple fuel purchases.

We group by trip id, and sum the costs.

3. Merge the trips, loads and aggregated fuel costs in a master_df.csv. (using .merge) Master_df looks like this:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	trip_id	load_id	driver_id	truck_id	trailer_id	dispatch_date	actual_distance_miles	actual_duration_hours	fuel_gallons_used	average_mpg	idle_time_hours	trip_status	customer_id	route_id	total_revenue	load_type	actual_fuel_cost	net_trip_margin	revenue_per_mile
2	TRIP00000001	LOAD000000001	DRV00117	TRK00035	TRL00187	01-01-2022	1314	26.2	189.8	7.15	3.5	Completed	CUST00183	RTE00019	3551.95	Dry Van	779.45	2772.5	5.703158295281583
3	TRIP00000002	LOAD000000002	DRV00141	TRK00108	TRL00082	01-01-2022	515	8.6	93.6	5.8	8.3	Completed	CUST00076	RTE00058	1323.09	Dry Van	752.7	570.3899999999999	2.569106796116505
4	TRIP00000003	LOAD000000003	DRV00032	TRK00138	TRL00138	01-01-2022	2959	45	336.1	7.4	12	Completed	CUST00027	RTE00048	7964	Refrigerated	3796.39	4167.610000000001	5.1741729772617857
5	TRIP00000004	LOAD000000004	DRV00063	TRK00105	TRL00018	01-01-2022	717	11.1	119.3	6.5	9.8	Completed	CUST00088	RTE00013	1499.53	Refrigerated	0	1499.53	2.0913947
6	TRIP00000005	LOAD000000005	DRV00044	TRK00076	TRL00054	01-01-2022	2243	35	326.9	6.82	11.6	Completed	CUST00185	RTE00020	4055.66	Dry Van	512.53	3543.13	1.808140882748322
7	TRIP00000006	LOAD000000006	DRV00120	TRK00062	TRL00007	01-01-2022	1977	39.4	358.8	5.51	6.6	Completed	CUST00193	RTE00004	4041.01	Refrigerated	3311.07	729.94	2.0440111279716744
8	TRIP00000007	LOAD000000007	DRV00055	TRK00002	TRL00093	01-01-2022	1350	20.9	230.8	5.85	2.3	Completed	CUST00189	RTE00038	4337.1	Dry Van	402.34	3934.76	5.212666666666667
9	TRIP00000008	LOAD000000008	DRV00085	TRK00148	TRL00148	01-01-2022	2748	50.3	437.6	6.28	10.5	Completed	CUST00049	RTE00016	6478.83	Refrigerated	581.66	5897.17	2.3576525384279474
10	TRIP00000009	LOAD000000009	DRV00129	TRK00014	TRL00083	01-01-2022	3226	51.4	489.5	9.93	4.1	Completed	CUST00079	RTE00025	5474.8	Refrigerated	0	5474.8	2.6970861748295103
11	TRIP00000010	LOAD000000010	DRV00039	TRK00116	TRL00126	01-01-2022	3388	60.2	460.3	7.36	5.7	Completed	CUST00079	RTE00025	6338.77	Refrigerated	0	6338.77	1.870947482
12	TRIP00000011	LOAD000000011	DRV00132	TRK00085	TRL00083	01-01-2022	538	9.9	72.2	7.45	4.9	Completed	CUST00127	RTE00058	1305.1999999999998	Refrigerated	499.18	806.0199999999998	2.4260223048327134

Merge the actual fuel costs, 0 cost if no fuel bought on trip

4. Next feature is we find Trip Margin and Revenue per Mile (RPM). Revenue per mile is equal to difference between total revenue and actual fuel cost

Then finally master_data.csv is created.

Block 4:

In the fourth block, we load the missing dimension tables if not already loaded.

We load trucks.csv, drivers.csv and customers.csv

trucks.csv looks like this:

	A	B	C	D	E	F	G	H	I	J	K
1	truck_id	unit_number	make	model_year	vin	acquisition_date	acquisition_mileage	fuel_type	tank_capacity_gallons	status	home_terminal
2	TRK00001	3463	Peterbilt	2016	1VV205190335317039	27-04-2017	18814	Diesel	200	Active	Omaha
3	TRK00002	6461	Kenworth	2015	1NV753749606229960	04-10-2018	26795	Diesel	150	Active	Seattle
4	TRK00003	2335	Peterbilt	2018	1NT803101860493229	17-09-2020	11795	Diesel	150	Maintenance	Atlanta
5	TRK00004	3317	Freightliner	2015	1VV131945381692351	17-06-2017	8875	Diesel	200	Active	Salt Lake City
6	TRK00005	6731	Freightliner	2016	1FT822215709701985	08-10-2021	42042	Diesel	250	Maintenance	Miami
7	TRK00006	6082	Kenworth	2017	1VV963516965177788	12-05-2018	47794	Diesel	150	Active	Minneapolis
8	TRK00007	7421	Volvo	2019	1MV134100489809236	29-01-2020	42328	Diesel	200	Active	Charlotte
9	TRK00008	3112	Mack	2015	1MV426140684859204	27-01-2015	27548	Diesel	150	Active	Houston

drivers.csv looks like this:

	A	B	C	D	E	F	G	H	I	J	K	L
1	driver_id	first_name	last_name	hire_date	termination_date	license_number	license_state	date_of_birth	home_terminal	employment_status	cdl_class	years_experience
2	DRV00001	Jennifer	Hernandez	31-10-2014		DL673510887	WA	07-11-1973	Denver	Active	A	3
3	DRV00002	William	Martin	02-10-2020		DL128955006	GA	03-11-1976	Columbus	Active	A	20
4	DRV00003	Charles	Hernandez	21-09-2021		DL523076025	NC	06-04-1970	Salt Lake City	Active	A	19
5	DRV00004	Barbara	Brown	08-09-2013		DL735540030	WA	06-02-1995	Denver	Active	A	19
6	DRV00005	Mary	Martinez	02-12-2018		DL706011277	AZ	15-07-1960	Chicago	Active	A	12
7	DRV00006	Robert	Taylor	04-04-2019		DL811890569	NC	02-05-1990	Portland	Active	A	23
8	DRV00007	James	Martinez	02-07-2019	17-05-2021	DL129100607	PA	23-05-1974	Miami	Terminated	A	5
9	DRV00008	Barbara	Wilson	09-06-2020	23-01-2021	DL190137896	CO	28-12-1978	Philadelphia	Terminated	A	15
10	DRV00009	Karen	Hernandez	29-09-2013		DL790147183	FL	22-02-1989	Los Angeles	Active	A	13

customers.csv looks like this:

	A	B	C	D	E	F	G	H
1	customer_id	customer_name	customer_type	credit_terms_days	primary_freight_type	account_status	contract_start_date	annual_revenue_potential
2	CUST00001	Metro Wholesale	Dedicated	60	General	Inactive	20-02-2020	985117
3	CUST00002	National Retail	Contract	30	Retail	Active	02-06-2021	4936566
4	CUST00003	XYZ Industries	Contract	30	Consumer Goods	Active	04-09-2020	3102814
5	CUST00004	American Corp	Contract	15	Food/Beverage	Active	29-11-2020	3948027
6	CUST00005	American Distribution	Spot	45	Consumer Goods	Active	04-02-2020	3682564
7	CUST00006	American Industries	Dedicated	30	Consumer Goods	Active	16-02-2021	973274
8	CUST00007	Superior Distribution	Spot	30	Food/Beverage	Active	05-02-2021	1499297
9	CUST00008	American Logistics	Contract	30	Automotive	Inactive	23-07-2021	1613275

Then we create dim_trucks.csv which has features like total maintenance cost, downtime hours, maintenance events count etc, dim_trucks.csv looks like this:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	truck_id	unit_number	make	model_year	vin	acquisition_date	acquisition_mileage	fuel_type	tank_capacity_gallons	status	home_terminal	total_maintenance_cost	total_downtime_hours	maintenance_events_count
2	TRK00001	3463	Peterbilt	2016	1VV205190335317039	27-04-2017	18814	Diesel	200	Active	Omaha	51775.99	713.2	24
3	TRK00002	6461	Kenworth	2015	1NV753749606229960	04-10-2018	26795	Diesel	150	Active	Seattle	54869.04	685.3	27
4	TRK00003	2335	Peterbilt	2018	1NT803101860493229	17-09-2020	11795	Diesel	150	Maintenance	Atlanta	90161.42	1133.1	41
5	TRK00004	3317	Freightliner	2015	1VV131945381692351	17-06-2017	8875	Diesel	200	Active	Salt Lake City	38561.82	483.2	22
6	TRK00005	6731	Freightliner	2016	1FT822215709701985	08-10-2021	42042	Diesel	250	Maintenance	Miami	66088.45	750.6	33
7	TRK00006	6082	Kenworth	2017	1VV963516965177788	12-05-2018	47794	Diesel	150	Active	Minneapolis	50780.83	587.7	26
8	TRK00007	7421	Volvo	2019	1MV134100489809236	29-01-2020	42328	Diesel	200	Active	Charlotte	25868.97	415.7	20
9	TRK00008	3112	Mack	2015	1MV426140684859204	27-01-2015	27548	Diesel	150	Active	Houston	33024.74	483	19

Block 5:

Then we create dim_drivers.csv which has features like 'driver_id', 'years_experience', 'tenure_years', 'employment_status' etc, dim_drivers.csv looks like this:

	A	B	C	D	E	F	G	H	I	J	K	L	M	
1	driver_id	first_name	last_name	hire_date	termination_date	license_number	license_state	date_of_birth	home_terminal	employment_status	cdl_class	years_experience	tenure_years	
2	DRV00001	Jennifer	Hernandez	31-10-2014		DL673510887	WA	07-11-1973	Denver	Active	A		3	10.17
3	DRV00002	William	Martin	02-10-2020		DL128955006	GA	03-11-1976	Columbus	Active	A		20	4.25
4	DRV00003	Charles	Hernandez	21-09-2021		DL523076025	NC	06-04-1970	Salt Lake City	Active	A		19	3.28
5	DRV00004	Barbara	Brown	08-09-2013		DL735540030	WA	06-02-1995	Denver	Active	A		19	11.31
6	DRV00005	Mary	Martinez	02-12-2018		DL706011277	AZ	15-07-1960	Chicago	Active	A		12	6.08
7	DRV00006	Robert	Taylor	04-04-2019		DL811890569	NC	02-05-1990	Portland	Active	A		23	5.74
8	DRV00007	James	Martinez	02-07-2019	17-05-2021	DL129100607	PA	23-05-1974	Miami	Terminated	A		5	1.88
9	DRV00008	Barbara	Wilson	09-06-2020	23-01-2021	DL190137896	CO	28-12-1978	Philadelphia	Terminated	A		15	0.62

Create dim_customers.csv, and it looks like this

	A	B	C	D	E	F	G	H
1	customer_id	customer_name	customer_type	credit_terms_days	primary_freight_type	account_status	contract_start_date	annual_revenue_potential
2	CUST00001	Metro Wholesale	Dedicated	60	General	Inactive	20-02-2020	985117
3	CUST00002	National Retail	Contract	30	Retail	Active	02-06-2021	4936566
4	CUST00003	XYZ Industries	Contract	30	Consumer Goods	Active	04-09-2020	3102814
5	CUST00004	American Corp	Contract	15	Food/Beverage	Active	29-11-2020	3948027
6	CUST00005	American Distribution	Spot	45	Consumer Goods	Active	04-02-2020	3682564
7	CUST00006	American Industries	Dedicated	30	Consumer Goods	Active	16-02-2021	973274
8	CUST00007	Superior Distribution	Spot	30	Food/Beverage	Active	05-02-2021	1499297
9	CUST00008	American Logistics	Contract	30	Automotive	Inactive	23-07-2021	1613275

Block 6:

Flagging Anomalies in this Block.

1. Flag MPG Anomalies (Heavy Duty Diesel Trucks typically get 5.0 - 8.5 MPG)
2. Calculate Idle Time Severity, # High idle time burns fuel without moving. Let's create a critical flag if idle time > 30% of total duration.

Master_df looks like this:

	A	B	C	D	E	F	G	H	I	J	K
1	trip_id	load_id	driver_id	truck_id	trailer_id	dispatch_date	actual_distance_miles	actual_duration_hours	fuel_gallons_used	average_mpg	idle_time_hours
2	TRIP00000001	LOAD00000001	DRV00117	TRK00035	TRL00167	01-01-2022	1314	26.2	183.8	7.15	3.5
3	TRIP00000002	LOAD00000002	DRV00141	TRK00108	TRL00082	01-01-2022	515	8.6	93.6	5.5	8.3
4	TRIP00000003	LOAD00000003	DRV00032	TRK00031	TRL00138	01-01-2022	2509	45	339.1	7.4	12
5	TRIP00000004	LOAD00000004	DRV00083	TRK00105	TRL00018	01-01-2022	717	11.1	110.3	6.5	9.6
6	TRIP00000005	LOAD00000005	DRV00044	TRK00076	TRL00054	01-01-2022	2243	35	328.9	6.82	11.6
7	TRIP00000006	LOAD00000006	DRV00120	TRK00062	TRL00007	01-01-2022	1977	39.4	358.8	5.51	6.6
8	TRIP00000007	LOAD00000007	DRV00055	TRK00002	TRL00093	01-01-2022	1350	20.9	230.8	5.85	2.3
9	TRIP00000008	LOAD00000008	DRV00065	TRK00035	TRL00148	01-01-2022	2748	50.5	437.6	6.28	10.5
10	TRIP00000009	LOAD00000009	DRV00129	TRK00014	TRL00083	01-01-2022	3226	51.4	465.5	6.93	4.1
11	TRIP00000010	LOAD00000010	DRV00029	TRK00116	TRL00126	01-01-2022	3388	60.2	460.3	7.36	5.7
12	TRIP00000011	LOAD00000011	DRV00132	TRK00085	TRL00083	01-01-2022	538	9.9	72.2	7.45	4.9

Block 7:

Create Final Fact Table (Fact_Trips)

Keep only the robust analytical columns to keep the file clean and performance oriented.

Fact_Trips.csv looks like this:

	A	B	C	D	E	F	G	H	I
1	trip_id	load_id	driver_id	truck_id	route_id	customer_id	dispatch_date	actual_distance_miles	actual_duration_hours
2	TRIP00000001	LOAD00000001	DRV00117	TRK00035	RTE00019	CUST00183	01-01-2022	1314	26.2
3	TRIP00000002	LOAD00000002	DRV00141	TRK00108	RTE00058	CUST00076	01-01-2022	515	8.6
4	TRIP00000003	LOAD00000003	DRV00032	TRK00031	RTE00048	CUST00027	01-01-2022	2509	45
5	TRIP00000004	LOAD00000004	DRV00083	TRK00105	RTE00013	CUST00088	01-01-2022	717	11.1
6	TRIP00000005	LOAD00000005	DRV00044	TRK00076	RTE00020	CUST00185	01-01-2022	2243	35
7	TRIP00000006	LOAD00000006	DRV00120	TRK00062	RTE00004	CUST00193	01-01-2022	1977	39.4
8	TRIP00000007	LOAD00000007	DRV00055	TRK00002	RTE00038	CUST00169	01-01-2022	1350	20.9
9	TRIP00000008	LOAD00000008	DRV00065	TRK00035	RTE00016	CUST00049	01-01-2022	2748	50.5

J	K	L	M	N	O	P	Q	R
fuel_gallons_used	average_mpg	idle_time_hours	total_revenue	actual_fuel_cost	net_trip_margin	revenue_per_mile	flag_anomaly_mpg	flag_excessive_idle
183.8	7.15	3.5	3551.95	779.45	2772.5	2.703158295	0	0
93.6	5.5	8.3	1323.09	752.7	570.39	2.569106796	0	1
339.1	7.4	12	7964	3796.39	4167.61	3.174172977	0	0
110.3	6.5	9.6	1499.53	0	1499.53	2.0913947	0	1
328.9	6.82	11.6	4055.66	512.53	3543.13	1.808140883	0	1
358.8	5.51	6.6	4041.01	3311.07	729.94	2.044011128	0	0
230.8	5.85	2.3	4337.1	402.34	3934.76	3.212666667	0	0
437.6	6.28	10.5	6478.83	581.66	5897.17	2.357652838	0	0

Phase 1 Complete: Full Enterprise Star Schema Exported Successfully.

Block 8:

Extract Date Components like “dispatch_year”, “dispatch_month”, “dispatch_day_of_week”, “dispatch_quarter” from “dispatch_date” in master_df.

Block 9:

Bring in booking_type and weight from loads_df